

CS 791: Quiz 6

1. Consider a discrete random variable $X \in \{1, 2, 3, 4\}$ with function $f(x) = x^2$

Let the target distribution P and proposal distribution Q be given by

x	$P(x)$	$Q(x)$
1	0.1	0.25
2	0.2	0.25
3	0.3	0.25
4	0.4	0.25

Suppose we generate the following samples:

$$S_P = [2, 3, 3, 4] \quad \text{from } P,$$

$$S_Q = [1, 2, 1, 4] \quad \text{from } Q.$$

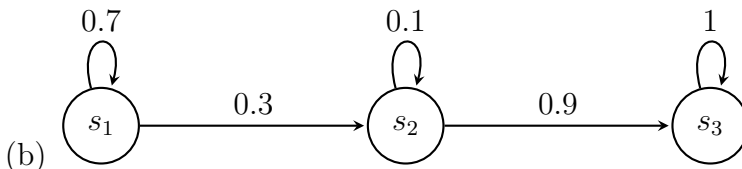
- (a) Compute $\hat{\mu}_f(S_P)$ using **regular sampling** with the samples S_P . ..1
 (b) Compute $\hat{\mu}_f(S_Q)$ using **importance sampling** with the samples S_Q . ..1

(a) $\hat{\mu}_f(S_P) = \frac{4+9+9+16}{4} = 9.5$

(b) $\hat{\mu}_f(S_Q) = \frac{1}{4}(1 \cdot \frac{0.1}{0.25} + 4 \cdot \frac{0.2}{0.25} + 1 \cdot \frac{0.1}{0.25} + 16 \cdot \frac{0.4}{0.25}) = 7.4$

2. Choose the correct option(s).

- (a) The burn-in phase is necessary in MCMC sampling to allow the MC to reach its stationary distribution.



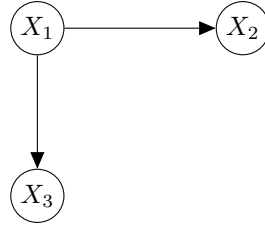
This MC is ergodic.

- (c) If $\pi(x)$ is the stationary distribution of an ergodic MDP with transition distribution T then $\pi(x)$ satisfies $\pi(x')T(x' \rightarrow x) = \pi(x)T(x \rightarrow x') \forall x, x' \in \chi$.
 (d) Gibbs sampling mixes poorly for highly correlated variables.

..2 (multicorrect, -1.0 for each wrong answer)

Correct: (a), (d)

3. Consider the Bayesian network with binary variables X_1, X_2, X_3 :



X_1	$P(X_1)$
0	0.4

X_1	X_2	$P(X_2 X_1)$
0	0	0.8
1	0	0.3

X_1	X_3	$P(X_3 X_1)$
0	0	0.9
1	0	0.5

The state is defined as (X_1, X_2, X_3) . Suppose we use the Gibbs transition function to sample points.

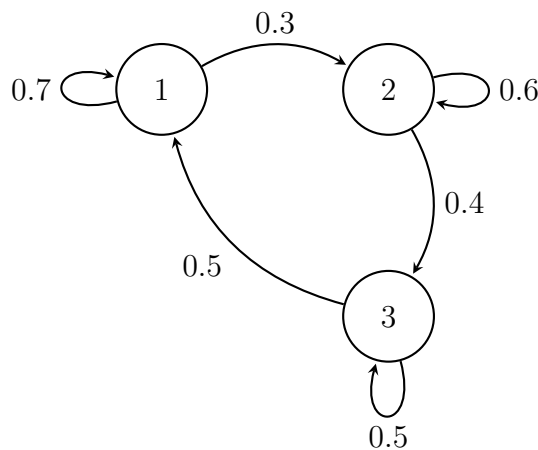
- (a) Find $3 \cdot T((1, 0, 1) \rightarrow (1, 1, 0))$..1
- (b) Find $3 \cdot T((1, 1, 1) \rightarrow (1, 0, 1))$..1
- (c) Find $3 \cdot T((1, 0, 1) \rightarrow (1, 0, 1))$..1

(a) $T((1, 0, 1) \rightarrow (1, 1, 0)) = 0$ (as the states differ in 2 X 's). So, the answer is 0.

(b) $T((1, 1, 1) \rightarrow (1, 0, 1)) = \frac{1}{3} \cdot \Pr(X_2 = 0 | X_1 = 1, X_3 = 1) = \frac{1}{3} \Pr(X_2 = 0 | X_1 = 1) = \frac{0.3}{3}$. So, the answer is 0.3.

(c) $T((1, 0, 1) \rightarrow (1, 0, 1)) = \frac{1}{3}(\Pr(X_1 = 1 | X_2 = 0, X_3 = 1) + \Pr(X_2 = 0 | X_1 = 1, X_3 = 1) + \Pr(X_3 = 1 | X_1 = 1, X_2 = 0)) = \frac{1}{3}(\Pr(X_1 = 1) + \Pr(X_2 = 0 | X_1 = 1) + \Pr(X_3 = 1 | X_1 = 1)) = \frac{1}{3} \cdot (0.6 + 0.3 + 0.5) = \frac{1.4}{3}$. So, the answer is 1.4.

4. Consider a 3-state MC with states $S = \{1, 2, 3\}$



Find the stationary distribution $\pi = (\pi(1), \pi(2), \pi(3))$.

- (a) What is the value of $\frac{\pi(2)}{\pi(1)}$? ..2
- (b) What is the value of $\frac{\pi(3)}{\pi(2)}$? ..2

(c) Compute the Metropolis-Hastings acceptance probability $A(1 \rightarrow 2)$. ..1

Solve these equations to get the answer:

$$\pi_1 = 0.7\pi_1 + 0.5\pi_3$$

$$\pi_2 = 0.6\pi_2 + 0.3\pi_1$$

$$\pi_3 = 0.5\pi_3 + 0.4\pi_2$$

(a) 0.75

(b) 0.8

(c) $A(1 \rightarrow 2) = \min[1, \frac{\pi(2) \cdot T(2 \rightarrow 1)}{\pi(1) \cdot T(1 \rightarrow 2)}] = 0$

Total: 12
